

Introduction:

1. Dataset

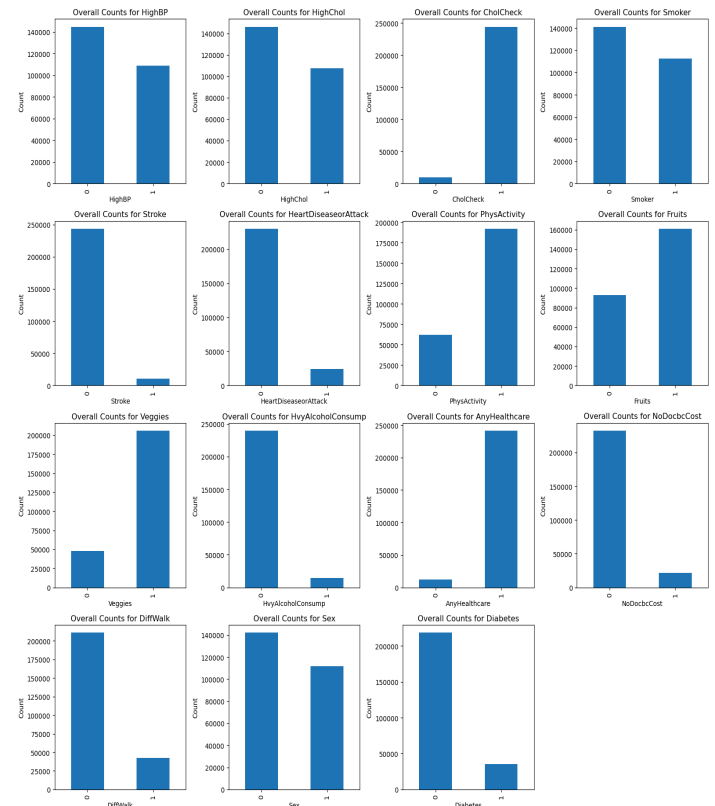
The CDC Diabetes Health Indicators dataset is designed to help researchers and analysts understand the prevalence of diabetes within the United States by providing health-related data from various demographic and geographic groups. The dataset contains a range of health indicators associated with diabetes, such as the percentage of individuals diagnosed with diabetes, age groups, gender, physical activity levels, and the availability of health services across different states and regions. This dataset is publicly available through the UCI Machine Learning Repository and is used extensively for classification tasks to predict the likelihood of diabetes diagnosis based on a variety of socio-economic and health-related factors. The dataset's primary objective is to aid in predicting diabetes trends and to support healthcare initiatives aimed at preventing and managing diabetes in diverse communities.

1.1 Identify database

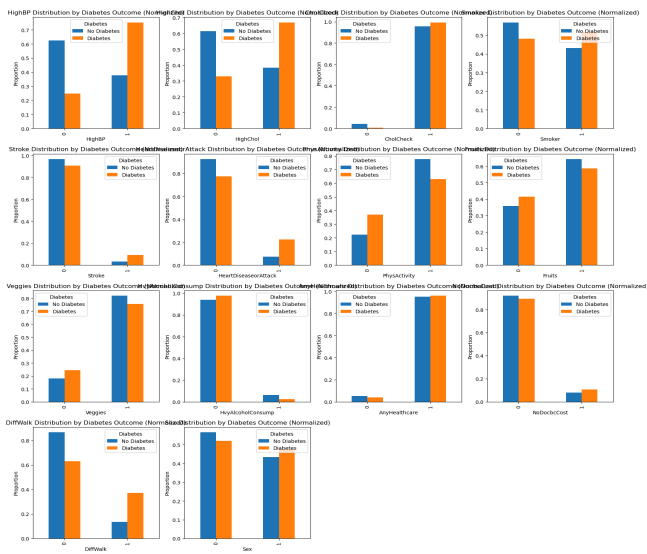
The dataset contains 254 thousand entries and consists of 21 features. The diabetes label had originally identified: no diabetes, pre-diabetes, and diabetes but for sake of simplification the pre-diabetes and diabetes label have been combined. All features are numerical with 14 being binary values and 7 being integer values within a limited range. The non-binary features are restricted and simplified to categorical groups representing ranges for survey responses, aside from BMI which is continuous. No values are missing and every entry is correctly formatted allowing for ease of use when it comes to exploring the data.

1.2 EDA

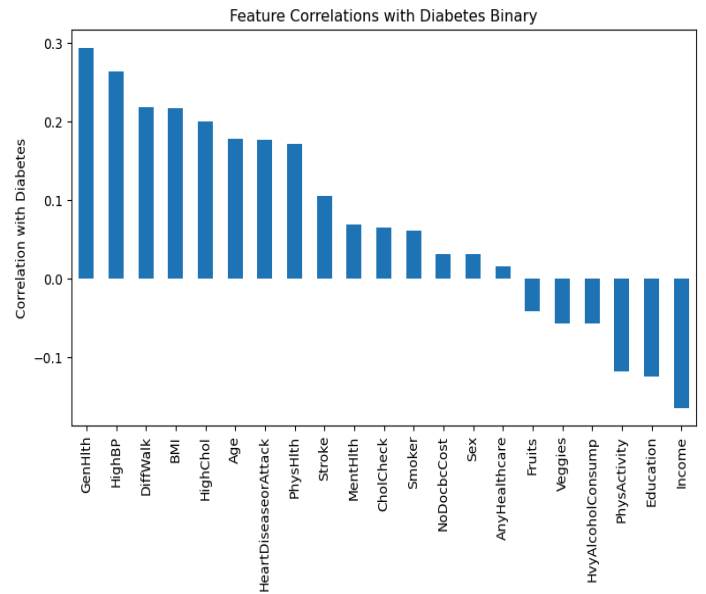
Examining the binary features helps visualize the type of survey responses that our models will take into consideration.



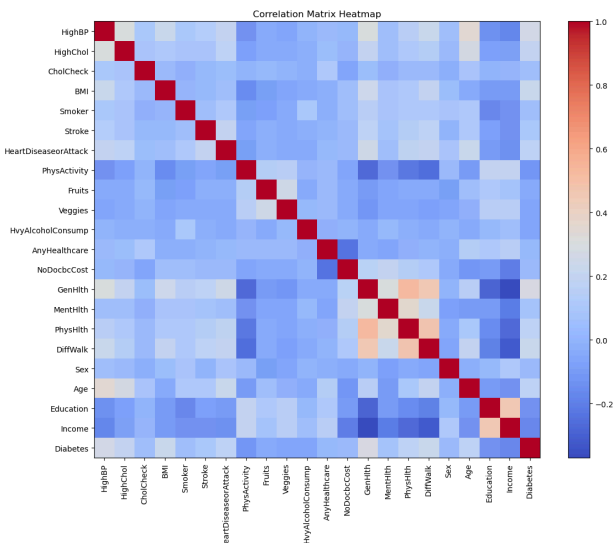
The entire dataset consists of 218,334 non diabetic entries and 35,346 diabetic entries. This indicates a heavily imbalanced dataset and will need to be accounted for in the creation of our models. The features with one sided responses such as cholesterol check and stroke are simple survey questions that would normally favor a certain response. Every other feature does not indicate any ambiguity within responses and may be useful. In order to evaluate these features it is best to observe a normalized ratio with respect to their diabetes label, to find any distinguishing factors.



The normalized distribution charts reveal key features for both labels. High blood pressure, high cholesterol, history of heart disease, and difficulty walking are all key features that distinguish a diabetic from a non diabetic. All other features have small differences which can still be used as a feature but may hurt our models due to their ambiguous nature.



Examining the remaining numerical features reveals that income, general health, and age, have distinguishable interquartile ranges. The other features indicate heavy outliers and similar IQR's, these features could be used effectively in a regularized model. Looking at a correlation map, we revealed possible features that can be combined such as income and education. Features with a higher correlation also showed high correlation to a diabetic label.



2. Predictive Task

Given the dataset we are aiming to evaluate: Is diabetes and prediabetes able to be predicted through the use of binary classification models, which are most effective, and does creating a new feature that includes labeling from a clustering algorithm improve evaluation metrics. The main baseline for our models will be a simple logistic regression model and random forest classifier as hyperparameter tuning can be easily computed and evaluated. The main models to be used for evaluation are SVM with L1 and L2 regularization with and without a

clustering label feature, a simple implementation of an extreme gradient boosting decision tree, and a catboost model. SVM models are heavily used when it comes to binary classification research allowing for easy evaluation of our features. These models have the advantage of adding regularization which can skew certain features or completely ignore them in order to improve accuracy. XGB is a relatively new and continuously improving classification model that may prove advantageous given the size of the dataset.

2.1 Evaluation

1. Random Forest Classifier

The Random Forest Classifier is an ensemble learning method that utilizes multiple decision trees to make predictions. It is capable of handling both regression and classification tasks, efficiently managing both numerical and categorical data. This model is particularly effective in capturing non-linear relationships within the data, while also providing robustness against overfitting by averaging the results of multiple trees. [ADDING MORE RELEVANT TO OUR RESEARCH QUESTION]

Hyperparameter Tuning: The Random Forest model underwent a hyperparameter optimization using GridSearchCV, which searched through a parameter grid consisting of the number of estimators, maximum depth, minimum samples required for splitting and leaf nodes, and whether bootstrap sampling is used. The best parameters found were:

```
n_estimators = 200
max_depth = 7
min_samples_split = 2
min_samples_leaf = 1
bootstrap = True
```

Results: The model achieved an accuracy of approximately 87%. However, it primarily excelled at identifying non-diabetic cases (precision = 0.87, recall = 0.99) but struggled significantly with identifying diabetic cases (precision = 0.60, recall = 0.06). This imbalance resulted in a low F1-score for the minority class (diabetic patients). While the Random Forest model performed well overall, the results indicate that there is room for improvement in terms of handling class imbalances in medical datasets.

2. Logistic Regression

Logistic Regression is a linear model used for binary classification tasks. It is known for its simplicity and efficiency in predicting probabilities that a sample belongs to a certain class. It works well when the relationship between the input features and the target variable is approximately linear, which when investigated proved true.

Hyperparameter Tuning: The hyperparameter tuning for Logistic Regression involved the regularization strength (C), and the penalty type (l1, l2). The best-performing parameters were:

```
C = 0.01
penalty = 'l1'
solver = 'liblinear'
```

Results: The model achieved an accuracy of 87%, similar to the Random Forest model. The precision for non-diabetic cases was high (0.88), while recall remained good (0.98). However, recall for the diabetic class was very low (0.16), and the F1-score was equally poor (0.25). This indicates that Logistic Regression, while fast and efficient, also suffers from the same issues of class imbalance. The model was efficient but not ideal for handling rare events in the dataset.

3. CatBoost

CatBoost is a gradient boosting algorithm specifically designed to handle both numerical and categorical variables without extensive preprocessing. It builds an ensemble of decision trees in a sequential manner, where each new tree seeks to reduce the residual errors of the previous trees. CatBoost also combats overfitting through permutations of the training data and gradient-based feature selection, often resulting in more stable and accurate performance compared to other boosting methods. In the context of our diabetes dataset, the ability to handle diverse feature types is especially beneficial, given the mix of continuous health indicators and potential categorical data like demographic factors.

Hyperparameter Tuning:

We applied GridSearchCV to find the optimal hyperparameters for CatBoost, exploring various configurations of learning rate, maximum tree depth, number of boosting iterations, and L2 regularization settings. Each parameter combination was assessed through 5-fold cross-validation using accuracy as the evaluation metric. The best set of parameters discovered was:

```
iterations = 100
depth = 7
learning_rate = 0.1
l2_leaf_reg = 1
```

These settings suggest that a moderately deep tree (depth=7) paired with a mid-range number of iterations (100) and a standard learning rate (0.1) is optimal for balancing learning speed and error reduction. The relatively low L2 regularization (1) helps control overfitting without heavily constraining the model.

Results:

Under these best-found hyperparameters, CatBoost achieved ~87% accuracy on the test set, similar to other models we tested. However, despite its strong performance on the majority class (non-diabetic), it exhibited a recall of only 16% for the minority class (diabetic). In practical terms, this means the model correctly identifies just 16% of diabetic patients while missing the remaining 84%. This shortfall is critical in a medical context, where failing to flag a true diabetic case poses significant health risks. The low recall for class 1 underlines the dataset's imbalance (roughly 43,739 non-diabetic vs. 6,997 diabetic samples) and indicates that further measures—such as class weighting, oversampling, or threshold tuning—are necessary to improve detection of the diabetic population.

4. Support Vector Machines (No Cluster Labels)

Support Vector Machines (SVMs) are robust classifiers that work by identifying the optimal hyperplane to separate classes in a high-dimensional feature space. In this evaluation, two SVM models were implemented, one with L2 regularization and another with L1 regularization.

Hyperparameter Tuning: Due to limitations of hardware the regularization parameter C was kept at a value of 1.0. These models took over 30 minutes to converge and exploring different values would not be feasible within a timely manner.

Results:

Both SVM models yielded identical classification reports, reflecting the challenges associated with our imbalanced medical dataset. The performance metrics are summarized below:

Metric	SVM L2	SVML1
Non Diabetic	Precision: 0.95 Recall: 0.72 F1: 0.82	Precision: 0.95 Recall: 0.72 F1: 0.82
Diabetic	Precision: 0.31 Recall: 0.78 F1: 0.44	Precision: 0.31 Recall: 0.78 F1: 0.44
Overall Accuracy	~73%	~73%

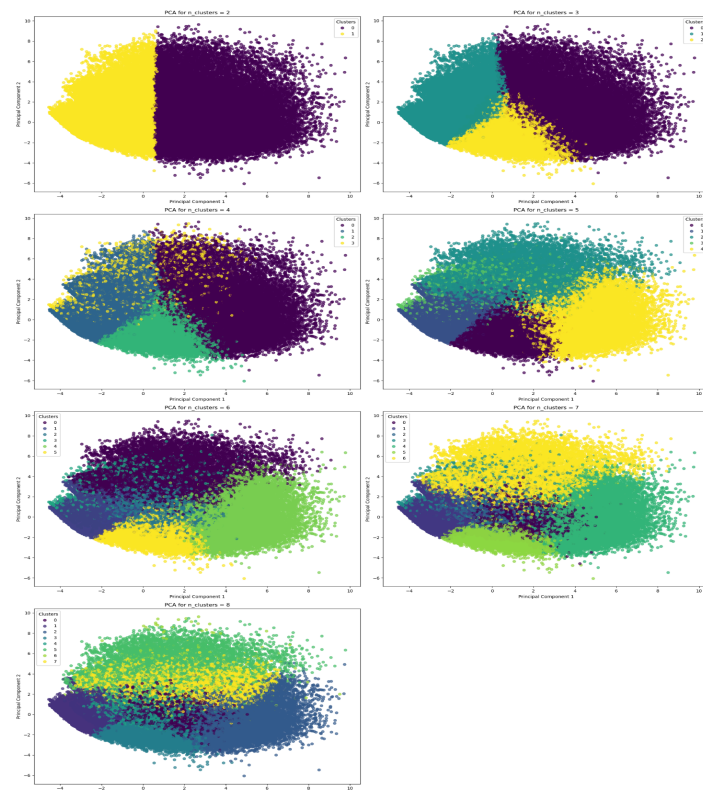
While both SVM classifiers perform comparably, they excel at identifying non-diabetic cases but struggle with accurately classifying the diabetic minority. These results underscore the challenges of working with imbalanced datasets in medical research and suggest that further strategies such as alternative tuning methods or class imbalance handling techniques may be needed to improve diabetic case detection. The low score reports on diabetic classification will be a common occurrence for future models.

Creating Cluster Labels

Since there are a lack of features within the original dataset it would be best to be able to find patterns that otherwise would go unnoticed. K-Means clustering is an unsupervised method which allows the creation of its own unique labels/groups. Using K-means clustering we would be able to specify the amount of groups within the data and possibly discover hidden patterns. The data originally had 3 labels, so 3 clusters would be enough to distinguish those groups; however four or more clusters may also reveal its own high dimensional pattern. With these new labels we can retrain the SVM models and possibly gain higher evaluation metrics.

In the figure below, the clusters with distinct groups under PCA outline two and three clusters as possible useful features. Four clusters may

also prove useful as it might be separable in a higher dimension but five and above shows too much overlap and may worsen our models. For sake of consistency the clearest defined clusters (n = 3) will be used for evaluation.



5. Support Vector Machines (With Cluster Labels)

In this evaluation, we extended the feature set by incorporating cluster labels derived from a prior clustering analysis to capture latent structures within the data. Two SVM models were implemented: one with L2 regularization and another with L1 regularization.

Hyperparameter:

The same restraints were met with the regular SVM models.

Results:

Both SVM classifiers, when augmented with cluster labels as features, produced identical

classification reports. The performance metrics are as follows:

Metric	SVM L2	SVM L1
Non Diabetic	Precision: 0.96 Recall: 0.72 F1: 0.82	Precision: 0.96 Recall: 0.72 F1: 0.82
Diabetic	Precision: 0.31 Recall: 0.79 F1: 0.44	Precision: 0.31 Recall: 0.79 F1: 0.44
Overall Accuracy	~73%	~73%

The consistency in these results indicates that, despite the introduction of cluster labels as additional features, the SVM classifiers continue to excel at identifying non-diabetic cases while facing challenges in accurately predicting the diabetic class. There was only a .01 increase in precision and no other metric changed indicating it being a negligible increase. Clusters could provide a benefit if the data was balanced.

6. XGB (Extreme Gradient Boost)

An XGBoost Classifier is a gradient boosting algorithm that builds multiple decision trees and optimizes predictive performance. It's built for non linear relationships and handling more structured data.

Hyperparameter Tuning:

XGB classifiers can take in a multitude of hyperparameters. The best found through A Grid Search are as follows:

Parameter	Value	Description
n_estimators	100	Training rounds
learning_rate	0.1	Step size moderator

max_depth	5	Tree depth
reg_lambda	1.5	L2 regularizer

Results:

The model achieved an overall accuracy of 87% however, the performance was uneven across both classes.

Metric	Non-Diabetic	Diabetic
Precision	0.88	0.57
Recall	0.98	0.16
F1-Score	0.93	0.25

This once again reinforces the unbalanced groups within the medical dataset but shows improvement over a normal decision tree. XGB relies on high volume data and this dataset can provide that. It struggles with classifying the diabetic class however it showed improvements due to finding the non-linear relationships within the data.

3 Conclusions

Out of every model in comparison to the baseline logit and random forest models, the SVM models with cluster labeling had an overall increase across all evaluation metrics regarding the diabetic class. Logistic regression has extremely high metrics regarding the non-diabetic class. But very low metrics when it comes to the diabetic class, this would be the same as guessing non-diabetic each time. The accuracy results to 87 percent as that is close to the actual percentage of non-diabetic to diabetic labels with respect to the dataset. So any increase in performance is shown through an increase in metrics across the diabetic class. The SVM model has a 394% increase in recall performance and 76% higher F1-score for

diabetic cases. This comes at the cost of 19% decrease in overall accuracy.

The most promising approach emerged with the SVM models, especially when augmented with cluster labels derived from K-Means. These additional features provided a slight boost in capturing the diabetic class, although the improvement was not large. The uniformly low recall scores for diabetic patients across all models emphasize the significant challenge of class imbalance in medical datasets. For future work, techniques such as class weighting, oversampling (e.g., SMOTE), or adjusting probability thresholds may be essential to increase the detection rate of true diabetic cases and thus make these models more clinically reliable.

Dataset Source:

UCI Machine Learning Repository. (n.d.). *CDC Diabetes Health Indicators Data Set* [Data set]. Retrieved [March,10,2025], from <https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators> doi:10.24432/C53919

Github

https://github.com/outsider1113/dsc148_proj